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Original article

Effects of high and low volume of strength training on muscle strength, muscle volume and lipid profile in postmenopausal women

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Abstract

Changes in lipid profile are considered a risk factor for cardiovascular disease (CVD), especially in postmenopausal woman who have been associated with age-related loss of muscle mass. The beneficial role of aerobic exercise in the prevention of CVD has been well documented. However, the effect of strength training has not been established. The purpose of this study was to determine the changes of lipoprotein levels after 12 weeks of different volumes of strength training and its correlation with strength and muscle volume in postmenopausal women. The participants were randomized into three groups: low volume (LVST; $n = 12$, 1 set) and high volume of strength training (HVST; $n = 11$, 3 sets), or control group ($n = 12$). Training groups performed 12 weeks of supervised strength exercises, 15 maximum repetitions, five times a week, 20 minutes for LVST and 40 minutes for HVST for each training session. Measurements included body composition, strength and muscle volume, as well as blood analysis (glucose, total cholesterol, triglycerides, low-density lipoprotein, and high-density lipoprotein) pre- and post-training. The HVST and LVST improved the one-repetition maximum knee extension strength ($p < 0.001$), maximal dynamic strength ($p < 0.001$), and muscle volume ($p = 0.048$). Post-training triglyceride was lower in HVST when compared to LVST and the control group ($p = 0.047$). Even though they present the same neuromuscular and morphological adaptations in postmenopausal women, the HVST is more effective than LVST in improving the lipid profile of postmenopausal woman, and can be considered as an ideal program of intervention to reverse changes in lipid metabolism commonly found in this group.
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Keywords: Aging; Menopause; Muscle strength; Muscle thickness; Strength exercise; Triglycerides

Introduction

Hyperlipidemia is a well-documented risk factor for cardiovascular disease (CVD), and is the major cause of death in men and women,[1–3] especially in postmenopausal women. Increased risk of CVD in menopausal women has been associated with age-related muscle mass loss, changes in body composition, fat deposition, and functional capacity.[2,4]

Previous studies demonstrated that there is a significant contribution of the regular practice of aerobic exercise in the prevention and control of CVD.[2,5–8] However, little is known about the preventive action of strength training on the major risk factors for developing of CVD, such as hypertension, cholesterol levels, obesity, and type 2-diabetes.[3–8]

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Strength training has been recommended as a non-pharmacological strategy to decrease serum concentrations of total cholesterol (TC), triglycerides (TAG), and low-density lipoprotein (LDL), and raise high-density lipoprotein (HDL) concentration. In addition, strength training increases insulin sensitivity,[2,8] decreases plasma TAG fasting,[9] although, some authors have reported no changes in lipids and lipoproteins levels after a strength training program.[2,5−7,10]

Methodological differences in previous findings may be due to differences in the duration of training, types of strength training (e.g., circuit-machines, free-weights, or elastic bands), intensity, and mostly total resistance exercise volume. The volume of training is a product of number sets performed for each exercise[11] and total training volume plays a significant role in muscle volume (MV) and blood lipid profile changes.[12]

However, the optimal number of sets (1 vs. 3) or low versus high volume of the exercises to develop MV remains controversial. The loss of MV leads to substantial decline in functional capacity, an increased risk of falls, fractures, and developing of chronic metabolic diseases followed by CVD.[4]

Strength training has been shown to be an effective intervention for body composition, mainly decreasing body mass and body fat, and increasing muscle mass. However, the impact of resistance training on lipid profile remains unclear and requires further investigation.[1−3,13,14] With respect to the lipid profile, the controversy among the studies is greater because few studies have investigated the effect of strength training in lipids and lipoproteins and the studies are contradictory.

Hormonal changes (i.e., decrease of estrogen) in postmenopausal women result in a loss of the cardioprotective effect of endogenous estradiol and their incidence of cardiovascular disease rises above that of men.[7,10] We have not found other works that evaluate the effect of strength training on MV and concentration of lipoproteins in postmenopausal women. To the authors' knowledge no study has yet examined the effects of low volume strength training (LVST) versus high volume strength training (HVST) on lipoproteins in postmenopausal women and the relation of this with MV.

Therefore, the purpose of this study was to determine the responses of lipid and lipoprotein cholesterol concentrations and the relation with strength and MV after 12 weeks of different volumes of strength training in postmenopausal women.

Materials and methods

Participants

Thirty-five healthy postmenopausal women [age, 59.5 ± 6.3 years; body mass, 68.2 ± 11.3 kg; height, 158.9 ± 7.2 cm; body mass index (BMI), 25.7 ± 8.8 kg/m^2; waist circumference, 79.8 ± 8.7 cm] were recruited; they were not engaged in regular and systematic strength training for at least 1 year prior to the study. The study excluded individuals with history of severe endocrine, metabolic, cardiovascular disease, neuromuscular diseases, diabetes, dyslipidemia, and nonsmokers. Participants were recruited through advertisement in a widely read local newspaper. Prior to the study, participants were carefully informed of the design, especially the possible risks and discomforts related to the procedures. Each selected volunteer was informed about the methodological procedures of the study and agreed to participate by signing an informed consent document, which was approved by the Federal University of Rio Grande do Sul Institutional Ethics and Research Committee.

Study design

This was a longitudinal study that lasted for 12 weeks of training. Forty volunteers were divided randomly, because they did not show any significant difference in basal concentrations of lipoproteins and maximum dynamic strength (1RM), into LVST ($n = 12$, 1 set), HVST ($n = 11$, 3 sets), and control group (CG = 12), which did not perform physical training during 12 weeks of study. The groups LVST and HVST performed eight exercises (bench press, biceps curl, triceps halter, 1 arm row back, leg press, knee extensor, knee flexion, and abdomen crunch), all exercises performed in 15 maximum repetitions (RM); a 40-second time interval was used between sets and exercises. The duration of each training session was approximately 20 minutes for LVST and 40 minutes for HVST. One week prior to the first evaluation, all participants had their lipid profile analyzed, and 1 week prior to the first evaluation, all participants were familiarized to strength training exercises, performed one RM (1RM) test on a knee extensor exercise machine at two different times. During the 12 weeks of training, the participants were instructed nutritionally by professionals in the area, so that there were large differences in the diets of the participants. All variables of this study were analyzed pre- and post−strength-training program. The ambient temperature (22°C) and the relative air humidity (65%) were controlled in pre- and post-training.

Characterization of participants

Prior to the strength training program, participants attended the laboratory for a preliminary session where anthropometric data were collected. Body mass and height were recorded for the calculation of BMI [body mass (kg)/height (m^2)]. Waist circumference was measured at the point of least perimeter between the iliac crest and the last rib. Thigh circumference was measured at the point larger MV of the right thigh. The skinfolds were measured at seven sites (triceps, sub-scapular, pectoral, axillary, abdomen, suprailiac, and thigh) using a Lange caliper (Beta Technology Inc., Cambridge, MD, USA), according to the method recommended by the International Society for the Advancement of Kinanthropometry,[15] and were used to calculate the body fat percentage.[16]

Muscle strength evaluation

The knee unilateral extensor 1RM test of the dominant leg was performed through a knee extensor exercise with the same "extension chair" equipment used for the training sessions (Taurus, Porto Alegre, RS, Brazil; resolution 0.1 kg). To the

control of the movement speed during the test, a Quartz metronome with a 1 Hz resolution was used.[4] There were no significant differences between the 1RM tests at baseline (test 4.0 ± 3.1 kg; retest 14.5 ± 3.5 kg), and the intraclass correlation coefficient was 0.85.

MV evaluation

Morphologic changes due to training were evaluated through images obtained with a B-mode ultrasound device (Philips VMI; Indústria e Comércio Ltda. Lagoa Santa, MG, Brazil) with a 7.5-MHz transducer. A water-based gel was used to promote acoustic contact between the skin and the transducer, which was positioned on the *rectus femoris* muscle at two-thirds of the distance from the greater trochanter of the femur to the lateral epicondyle and 3 cm lateral from the limb midline.[17] The subcutaneous adipose tissue and bone tissue were identified on the ultrasound image, and the distance between them was defined as the muscle thickness. When thickness reliability was assessed, variations were <3%. All ultrasound images were acquired from the dominant limb of the participant by the same evaluator, and maps were developed to ensure that images from the three evaluations were obtained at the same location of the muscle belly. Finally, the knee extensor MV was obtained through the method described by Miyatani et al,[18] where the MV is calculated by taking into account the muscle thickness (MT) and the thigh length (TL) through the equation:

$$MV = (MT \times 320.6) + (TL \times 110.9) - 4437.9. \quad (1)$$

Blood samples and biochemistry

Blood samples were analyzed to determine the levels of glucose (GLU), HDL-cholesterol, LDL-cholesterol, TAG, and TC. Upon arrival at the laboratory, participants rested for 5 minutes whereas a baseline (12-hour overnight fasting) blood sample was collected in the antecubital vein region using 12-mL disposable syringes. Each blood sample collected corresponded to a volume of 10 mL. For analysis of lipids, blood was stored in tubes without anticoagulant, and blood samples for GLU analysis were stored in tubes with fluoride anticoagulant. These samples were centrifuged at 1400*g* at 4°C for 10 minutes and supernatants were frozen at −75°C for later analysis. The concentrations of all lipids were analyzed using the enzymatic colorimetric automatic method (Cobas c111 analyzer, Roche, Basel, Switzerland).

Statistical analysis

Data normality was determined by the Shapiro–Wilk exploratory test. Training effects on the absolute values of muscle strength, MV, and lipid profile were tested through a two-way analysis of variance (group vs. time), followed by a Bonferroni *post hoc* test. Sample size was calculated using G POWER software (version 3.0.1; Heinrich "Heine" Universität, Düsseldorf, Germany) and determined that a sample size of $n = 35$ would provide a statistical power > 0.85 for all variables. Additionally, relative changes (values normalized to the baseline value from each participant) were calculated to determine the training effects on the knee extensor strength, MV, and lipid profile. All tests were performed in SPSS version 19.0 (SPSS Inc., Chicago, IL, USA). A value of $p < 0.05$ was considered significant for all analyses.

Results

Participant characteristics are listed in Table 1. No significant differences existed between groups pre- and poststrength training for height, body mass, body composition, muscle mass, BMI, waist, thigh circumference, or waist–hip ratio for groups HVST, LVST, and CG.

Significant time effects were observed for strength 1RM ($p < 0.001$; Fig. 1A). Twelve weeks of strength training improved the absolute values of maximal dynamic strength and MV (Fig. 1B) for HVST and LVST ($p < 0.001$ and $p = 0.048$, respectively) compared with CG.

Table 2 shows that there were no significant differences between groups in lipid profile (GLU, TC, HDL, and LDL) after strength training were observed among groups. However, TAG in post training was significantly different HVST in response to training for LVST and CG ($p = 0.047$; Fig. 1C).

Discussion

The key finding of our study is that a HVST modifies circulating TAG concentration. The results show significant increases in 1RM strength for HVST and LVST in

Table 1
Participant characterization and metabolic response, pre- and post-12 weeks of strength training.

Variables	HVST, $n = 11$		LVST, $n = 12$		CG, $n = 12$	
	Pre	Post	Pre	Post	Pre	Post
Body mass (kg)	66.4 ± 8.8	65.3 ± 8.7	70.1 ± 7.9	67.6 ± 7.1	65.6 ± 6.01	62.6 ± 13.7
Body fat (%)	38.3 ± 4.6	37.3 ± 3.7	37.8 ± 5.1	36.5 ± 5.2	36.3 ± 5.1	34.1 ± 6.9
Muscle mass (%)	33.1 ± 4.1	35.9 ± 3.8	37.1 ± 3.8	38.1 ± 3.1	36.2 ± 2.8	35.4 ± 3.8
BMI (kg/m^2)	25.4 ± 4.1	25.2 ± 3.5	26.1 ± 3.6	26.5 ± 3.8	25.7 ± 2.6	26.8 ± 5.7
Waist circumference (cm)	78.6 ± 9.2	77.2 ± 12.2	81.4 ± 7.4	80.2 ± 5.7	79.8 ± 6.5	78.4 ± 14.2
Thigh circumference (cm)	50.2 ± 9.2	52.2 ± 4.2	53.4 ± 7.8	54.2 ± 7.7	50.8 ± 9.5	50.4 ± 7.6
Waist–hip ratio	0.79 ± 0.06	0.78 ± 0.11	0.77 ± 0.05	0.76 ± 0.04	0.80 ± 0.05	0.79 ± 0.06

BMI = body mass index; CG = control group; HVST = high-volume strength training; LVST = low-volume strength training.

postmenopausal women, resulting from muscle hypertrophy. These findings show those 12 weeks of strength training produced substantial improvements in muscle strength and hypertrophy in both groups (HVST and LVST; Fig. 1A and B, respectively). Perhaps the most important benefit of increased muscle strength for postmenopausal women is a reduced risk of falling and sustaining injury.[1]

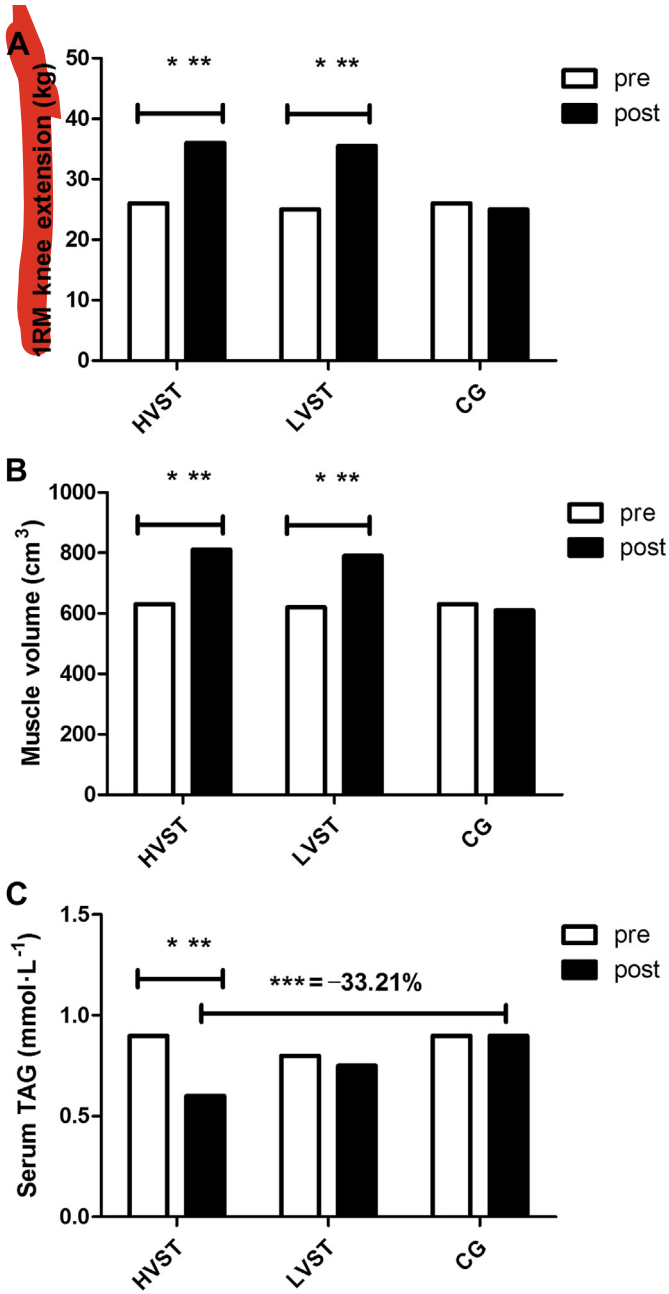

Fig. 1. (A) One-repetition maximum (1RM) knee extension strength, (B) muscle volume, and (C) triglyceride (TAG) concentrations in the fasted state before (pre) and after (post) 12 weeks of strength training in the high-volume strength training (HVST), low-volume strength training (LVRT), and control (CG) groups. There were no significant changes in these variables for CG. Values are mean ± standard deviation represented by vertical bars. * Significant difference between pre- and post-12 weeks of strength training ($p < 0.05$). ** Significant difference between HVST and LVST and CG ($p < 0.05$). *** Significant difference between HVST for LVST and CG ($p < 0.05$).

The different volumes of strength training tested in this study were ideally suited to postmenopausal women. By contrast, with our results, with respect to strength and neuromuscular adaptation, the literature demonstrates the superiority of the multiple-set (i.e., high volume).[11,19] It seems, according to the literature and our results, that HVST and LVST show the same adaptations in strength and hypertrophy. In this sense, the realization of LVST is sufficient for the development of strength and hypertrophy, because it has similar effects to the HVST, it is not necessary to the completion of high volume, which would increase the total time of the strength training session.

However, the results of this study for HVST showed a significant reduction in TAG, which did not occur in LVST (Fig. 1C); therefore, the realization of a greater volume of strength training should be primarily used by individuals with metabolic diseases such as obesity, type 2 diabetes, and metabolic syndrome, because HVST has an effect on the metabolism of fat. In fact, HVST should be recommended as a nonpharmacological strategy, considering its influence in reducing and attenuating several alterations involved in this period (sarcopenia, osteopenia, insulin resistance, inflammatory markers, among others).[13,20] However, HVST is still not routinely used as a therapeutic intervention to prevent and reverse the alterations induced by menopause[13] that increase TAG. In the present study, HVST and LVST show different results in TAG, with HVST significantly altering TAG after 12 weeks of training (Fig. 1C). This observation is different to results reported by Elliott et al[1] and Bemben et al.[21] Participants in those studies were hypertriglyceridemic and overweight. In accordance, Fahlman et al[10] observed a 22.5% reduction in TAG concentration in overweight but nonobese postmenopausal women with fasting TAG concentration similar to participants in the present study, 33.2% reduction of TAG (Fig. 1C). In agreement with our results, Tsekouras et al[22] reported a significant reduction in TAG following a single bout of whole-body isokinetic resistance exercise (12 exercises of 3 sets at 10 repetitions at 80% of maximum peak torque) in healthy, untrained men.

For the other parameters (GLU, TC, HDL, and LDL) the current training programs, both HVST and LVST, failed to evoke any significant alterations. Previous studies of strength training have not detected any difference in fasting TC,[19] HDL,[23] and LDL[24] concentrations between HVST and LVST groups and CG. Moreover, the bulk of evidence derived from the present research and other studies indicates that resistance exercise does not influence fasting TC, HDL, and LDL concentrations.

The HVST has a strong effect on TAG; this occurred because during a strength training session increases in energy demand are met with increases in the use of muscle and liver glycogen and intramuscular TAG,[7] resulting in glycogen and intramuscular TAG depletion.[25] Exercise is followed by a period of increased postexercise free fatty acid mobilization,[26] LPL activity,[27] and increased GLU uptake[19] to facilitate repletion of muscle glycogen and intramuscular TAG.[27] This phenomenon is closely linked to the duration of the training

Table 2
Lipid profile in fasting blood samples calculations for the three groups.

Fasting blood samples	HVST, $n = 11$		LVST, $n = 12$		CG, $n = 12$	
	Pre	Post	Pre	Post	Pre	Post
GLU (mM)	4.66 ± 1.53	4.98 ± 1.55	4.13 ± 1.96	3,87 ± 2.12	4.65 ± 1.5	4.75 ± 1.23
TC (mM)	5.36 ± 0.97	5.47 ± 0.86	6.01 ± 0.98	5.25 ± 1.08	5.12 ± 0.98	5.21 ± 0.97
HDL (mM)	1.70 ± 0.28	1.76 ± 0.38	1.61 ± 0.62	1.60 ± 0.62	1.58 ± 0.56	1.63 ± 0.23
LDL (mM)	3.21 ± 0.73	3.29 ± 0.63	2.87 ± 0.64	3.16 ± 0.75	3.23 ± 0.56	3.13 ± 0.47

CG = control group; GLU = glucose; HDL = high-density lipoprotein; HVST = high-volume strength training; LDL = low-density lipoprotein; LVST = low-volume strength training; TC = total cholesterol.

session, which was double in HVST (40 minutes) compared to LVST (20 minutes).

Our results are very important because they show that the higher difference between HVST and LVST (multiple-set vs. single-set) is the biochemical response (specifically in TAG), and there is no difference in neuromuscular and morphological response after 12 weeks of strength training in different volumes. Strength training produces an increased energy demand that is met by increased utilization of muscle and liver glycogen and intramuscular TAG,[7] thus resulting in glycogen and intramuscular TAG depletion.[1] Exercise is followed by a period of increased postexercise free fatty acid mobilization,[7] LPL activity, and increased GLU uptake[3] to facilitate repletion of muscle glycogen and intramuscular TAG.

Given that the reductions in TAG concentration were observed only in HVST, it is difficult to determine if these changes were exercise-induced or due to normal day-to-day variations. Because of this, a limitation of this study was that dietary consumption of the participants was not strictly monitored or recorded, but participants were instructed and consistently encouraged to maintain their normal diet during training and prior to blood collection.

In conclusion, the results of this study suggest that HVST can be used to reduce TAG after 12 weeks of intervention, although both volumes increase muscle strength and muscle thickness in postmenopausal women. HVST is more effective in decreasing the TAG concentration than LVST. For this reason, we recommend a program of strength training performed at high volume as a nonpharmacological intervention to prevent or even reverse the physiological changes induced by menopause.

Conflicts of interest

All authors declare no conflicts of interest.

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